VSLR

VSLR -

Development Plan

Product Overview

The VSLR product**, 'Naturoll'** will be a web-based card game where users collect and trade cards, partake in activities, and gain random cards.

- **Frontend:**

VSLR Naturoll will feature a modern browser interface, with responsive design.

- **Backend**:

A thorough backend will be constructed to ensure smooth and safe delivery of the site.

- **Database:**

Clean, well-organized databases will make the product run clean and simple.

- **Deliverables:**

-FIRST EDITION: APRIL 3RD

The first iteration of the product will have a barebones implementation of all the major systems. It will have the following features:

Acquiring random cards

Browsing user-owned cards

Generating personalized interface depending on user account.

Conducting trades to acquire new, specific cards.

Our Team

The VSLR team consists of Vihaan, Soreaksmey, London, and Ryan. We each have a particular role in the development of this project. Our team consists of developers with a wide scope of skills to manage specific sections which they specialize in.

- **Vihaan:**

Vihaan is taking the lead on developing the rewards management system and ensuring its high quality. The rewards system will be an integrated part of our game which will help users access a system where they can use the games virtual currency to gain more cards for their collection. I will also be responsible for handling the pack opening system within the game. Furthermore, I would be creating and achievement system where the player would be rewarded for meeting specific objectives (Like gaining their first 10 cards or gaining their first legendary). These achievements will also allow the player to gain the games virtual currency to further trade or obtain cards.

Deliverables

- Some sort of card gaining system where the user can gain a card at certain time interval/coins
- Quests which the user can complete daily to gain coins
- Virtual Currency which will be used for many functions of the game (Trading, Buying Cards, etc...)
- Achievement system

- **Ryan**:

Ryan is taking the lead on developing the inventory and data systems, which will be crucial.

The inventory system will keep track of what cards and other items users have and allow users to view and search through their cards.

Deliverables:

- A database of card information, including index numbers, names, pictures, rarity, stats, and descriptions.

- An inventory page showing all of your cards and any achievements you have, including a search bar and filters for rarity and card stats.
- A database system for storing which cards and achievements users have

- **Soreaksmey:**

Soreaksmey is taking the lead on developing the accounts and user systems, which will let different users access the site.

Deliverables

I think as the default, users can sign up and login via an email and password. I also want to give them a username so they can be more expressive. If the users prefer to sign up/login through a third-party account like Google, Facebook, or Apple, giving them that option would be good.

- **London:**

London is taking the lead on developing the trading system, and ensuring the transfer of cards works as intended

Deliverables – London

I plan to have a clean trading API with simple contracts to integrate with the rest of the platform. It should be as easy as calling RequestTrade(card, user), and awaiting a response. I will take responsibility for ensuring that transfer process is smooth and reliable.

Data Description

- **Accounts**

A table featuring all accounts, with unique identifiers for each user, and a link to their inventory

USER_ID	USERNAME	EMAIL	PREFS_ID	INV_ID	ACCESS
1	Playe1	email@email.com	12324122	12412531	Asof124yaiu
2	Garth	email@email.com	5675685	405340	Saduo124125asdywe
3	Soreaksmey	email@email.com	545235234	32349182	ADadohoy08407
4	Test	email@email.com	125235	60954683	Asdashfoas897142
5	Ryan	email@email.com	124124	123124	Qwroiyasdoi074124
6	5000	email@email.com	23523546	6547457	Asduhasoiuywe712409

- **Inventories**

A table linking ownership information to each card

CARD_ID	CARD_TYPE	NUM	OWNER_ID	TRADE_ID
1	123	-1	454276642	-1
2	43	87	468346	-1
3	427	-1	3456345634	-1
4	1874	-1	3465346436	46475734
5	853	-1	43643643634	-1

- **Trades**

A table demonstrating trades and their statuses.

HOST	SEND	GET	STATUS	TRADE_ID
Gary	1000c	cardID:6	PENDING	91230123
Jeanette	cardID: Dog	100c	REQUESTED (91230123)	112093108

Timmy	cardID: Cat	cardID: Horse	ACCEPTED	124124
Andrea	cardID: Horse	cardID: Cat	ACCEPTED	12412415

- **Awards**

A table storing awards and linking ownership information

AWARD	REQS_ID	OWNER_ID	STATUS	AWARD_ID
Ultimate Set	24135023	124087129837	Achieved	12412512
All Around the World	123124	80932598023	Pending	124352235

- **Cards**

A table storing the card information, and providing accessible unique identifiers.

CARD_ID	CARD_NAME	IMG	DESCRIPTION	RARITY TIER
1	Dog	https://url.com/dog	blablablabla	Legendary
2	Cat	https://url.com/cat	blablablabla	Uncommon
3	Horse	https://url.com/horse	blablablabla	Common
4	Pig	https://url.com/pig	blablablabla	Rare
5	Cow	https://url.com/cow	blablablabla	Elite
6	Rat	https://url.com/rat	blablablabla	Unseen

Functions Performed

- **Request Trade – Initialize new pending trade**

This will modify the table of Inventory and Trades. It will be capable of Inserting trades to the Trades data table, and assigning the Trade status to the inventory.

- **Confirm Trade – Confirms pending trade**

This will modify the table of Inventory and Trades. It is capable of deleting entries from the Trades data table, and modifying the Inventory data table.

- **Pull Cards – Get random cards**

This will modify the table of Inventory and Trades. It is capable of adding to the inventory table

- **Create Account – Initialize user account**

This will modify the table of Accounts. It is capable of adding to the accounts table.

- **Edit Settings – Edit user preferences**

This will modify the table of Accounts.

- **View Trades – Show cards on screen**

This will select from the trades table, the cards table, and the inventory table to determine which cards to show.

- **View Quests – Show rewards and quests on screen**

This will select from the cards table, inventory table, awards table.

Account Management

Summary The account management system is responsible for user authentication, profile, and providing APIs to link to the other systems.

- **Frontend:**

  User profile should be easily accessible, preferences should be able to be set from a GUI.

  Buttons should be easy to understand

- **Backend**:

  Link inventory references with user accounts, safely store passwords and preferences**.**

  Update user information from server.

- **Database:**

  SQL Database featuring securely stored accounts.

- **Deliverables:**

  -Account data structure

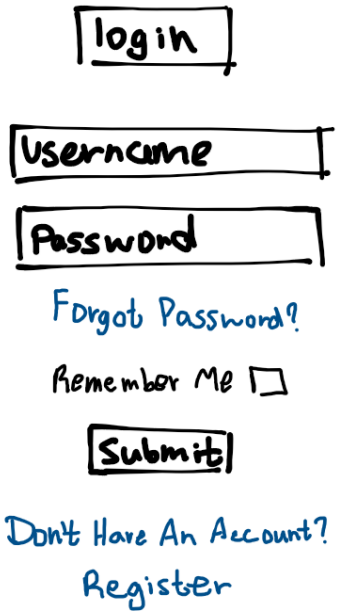
  -Verification

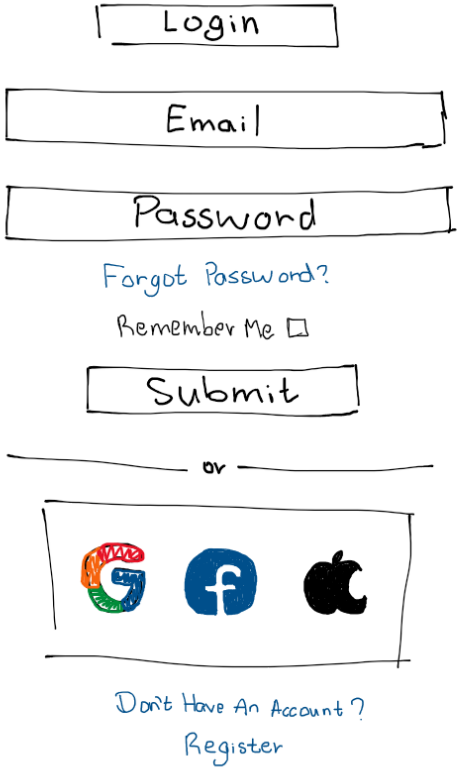
  -Let user have a username

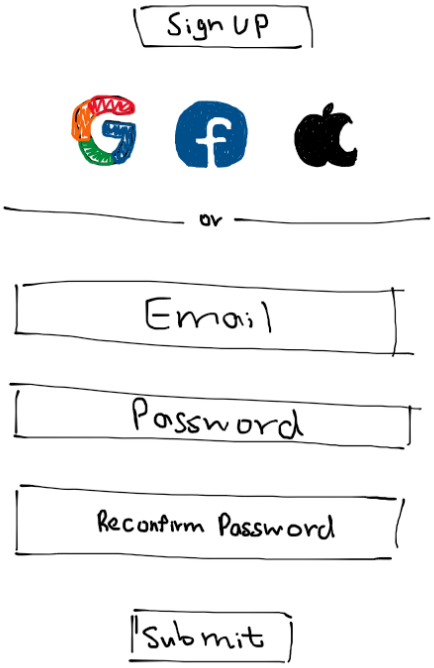
  -Login with email

  -Allows third party login

  -Login Screen:


login
Username
Password
Forgot Password?
Remember Me
Submit
Don't Have An Account?
Register


Login
Email
Password
Forgot Password?
Remember Me
Submit
or
Don't Have An Account?
Register


Sign UP
or
Email
Password
Reconfirm Password
Submit

Rewards Management

Summary The Rewards Management service is responsible for user engagement, such as daily activities, card pulls, and acquisition of cards.

- **Frontend:**

  Rewards Management should be front and center, quickly gathering user attention

- **Backend**:

  Update information in database, send requests for card acquisition. Automated loading rewards from data.

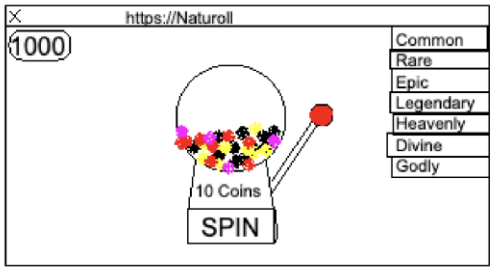
- **Database:**

  Easily definable rewards within database

- **Deliverables:**

  -Reward loading system

  -Reward interface



Inventory Management

Summary The Inventory Management service is responsible for handling data, including who has access to which cards, and how to manage ownership of cards.

- **Frontend:**

  Inventory Management should be simple on the frontend, and simply display the cards the user owns.

- **Backend**:

  Update information in database, handle control of ownership

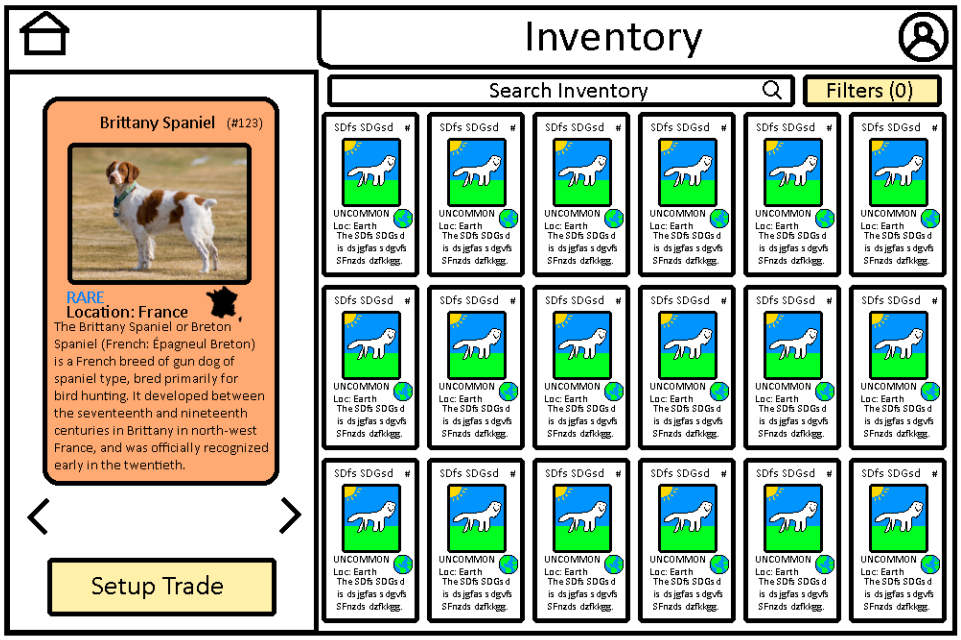
- **Database:**

  Easily definable cards within database

- **Deliverables:**

  -Card inventory and accessibility

  -Inventory API/interface



Trade Management

Summary The Trade Management service is responsible for handling user trades, ensuring access between inventory, other accounts, and asynchronous execution.

- **Frontend:**

  Trade management should display available trades, allow for selection of various trades in purchasing. Each trade must be confirmed by both users, and users must be able to post and configure trades to their liking using an easy interface.

- **Backend**:

  System of incoming and outgoing trade requests, it should run smoothly and perform its task with minimal hiccups or issues. The system must be able to match requests, and in close to real-time it should be able to update the database and remove trades that have been completed.

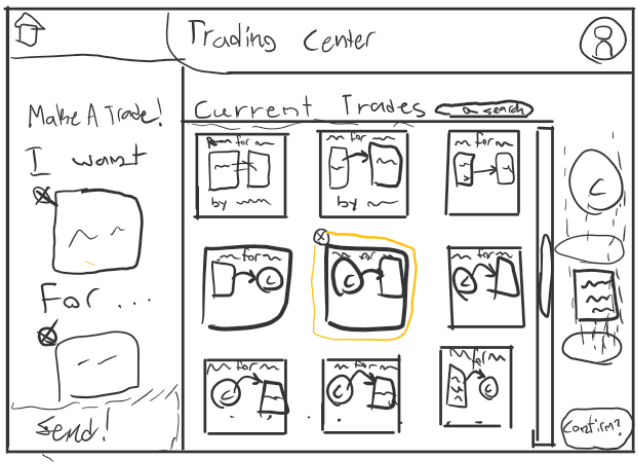
- **Database:**

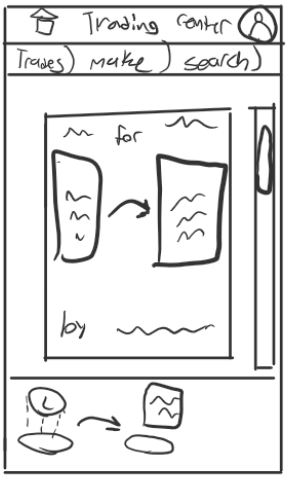
  Trades should be automatically generated and managed via automation, access should be optimized to allow for quick trading.

- **Deliverables:**

  -Trade viewport

  -Trade handling and exchange system





Wireframe by London

Client Analysis

Our client Tom Ding has the following requests of this service.

The game must be simple enough, yet have enough depth to justify continued playing. This aligns with our core vision of a simple framework that users are allowed to build onto, via collecting and trading cards. The simple addition of trades will add a whole new layer to the game, which will encourage users to collect and develop trading strategies to get their intended cards.

- **Industry Type:** Computer Science
- **Market Segmentation:** Gamers and Card Fans
- **Competition:** Nintendo

- **Client Feedback :**

Upon initial viewing of the documentation, our client found it detailed and solid, but in need of stronger product framing and execution details. To satisfy the client, we acknowledge their concern of a large MVP, and that a reduction in scale is optimal. In order to do this, we space out the different systems, and complete them incrementally. They are looking for reassurance on the security and safety of the site, which we can provide by doing additional research on best practices.